KG College of Arts and Science

(Autonomous) | Coimbatore

Student Attendance Monitoring and Eligibility Verification System

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Introduction

- Attendance plays a vital role in a student's academic performance. KG College of Arts and Science requires every student to maintain a minimum of 75% attendance to be eligible for semester examinations.

- The Student Attendance Monitoring and Eligibility Verification System is a Java-based application that automates the process of recording and managing student attendance.

- The system allows users to add student records, mark daily attendance, calculate attendance percentage, and instantly verify whether a student is eligible to appear for the examination.

- It is built using core Java concepts such as Object-Oriented Programming, ArrayList, and Scanner, making it simple, efficient, and easy to use.

- Manual attendance registers are time-consuming and prone to errors. This system eliminates such issues by providing instant and accurate results.

- The project is developed using Eclipse IDE and follows a clean menu-driven console interface that is easy to operate for both faculty and students.

Objectives

- To record and store student attendance data accurately.
- To calculate each student's attendance percentage automatically.
- To verify exam eligibility based on the 75% attendance rule.
- To allow adding new students and marking day-wise attendance.
- To provide a search feature to find individual student records.
- To generate a complete eligibility report for all students.

System Architecture

User Input

(Scanner)

Main Menu

(Switch-Case)

Student List

(ArrayList)

Student Class

(OOP Model)

Menu Options:

1. Add New Student
2. Mark Attendance
3. View All Students
4. Check Eligibility Report
5. Search Student

Student Class Methods:

- calculatePercentage() — computes attendance %
- isEligible() — checks if % ≥ 75
- markAttendance() — updates counts
- toString() — displays student row

Key Features

1. Add new students by entering their roll number, name, and initial attendance count.
2. Mark daily attendance as present or absent; the system updates counts automatically.
3. View a complete table of all students showing percentage and eligibility status.
4. Generate an eligibility report that lists who is eligible and who is not.
5. Search for any student by roll number or name to see their attendance details.
6. Duplicate roll numbers are rejected to maintain data accuracy.

Sample Output

Roll No.	Name	Total	Attended	%	Status
25BCS001	Prem Kumar	60	50	83.33%	ELIGIBLE
25BCS002	Mahapavithra	60	42	70.00%	NOT ELIGIBLE
25BCS003	Naresh Kumar	60	55	91.67%	ELIGIBLE
25BCS004	Adithya	60	38	63.33%	NOT ELIGIBLE

Result: 2 Eligible 2 Not Eligible Minimum required: 75%

Objectives of the Project

Accuracy	To eliminate manual errors in attendance recording and percentage calculation.
Automation	To automatically determine student eligibility without any manual computation.
Accessibility	To provide a simple interface that any faculty member can use without technical knowledge.
Efficiency	To save time by instantly generating attendance reports for all students.
Scalability	To support any number of students by using a dynamic ArrayList structure.

Working of the System

Step 1	The system starts and displays the main menu with five options.
Step 2	The administrator selects an option using the keyboard input.
Step 3	For new students, details such as roll number and name are entered.
Step 4	Attendance is marked day by day as present or absent for each student.
Step 5	The system calculates the percentage: (Attended / Total) × 100.
Step 6	If the percentage is 75% or above, the student is marked ELIGIBLE; otherwise NOT ELIGIBLE.

Conclusion

- The Student Attendance Monitoring and Eligibility Verification System successfully automates the attendance tracking process for college students using core Java programming.

- The system accurately records attendance, calculates percentages, and verifies eligibility based on the 75% rule, eliminating manual calculation errors completely.

- It provides useful features such as adding students, marking attendance, viewing reports, searching records, and generating an eligibility summary.

- The project demonstrates how Object-Oriented Programming, ArrayList, and Scanner can be effectively applied to solve real-world academic problems.

- In the future, this system can be enhanced by adding a graphical user interface, database connectivity, login authentication, and a web-based portal for students and faculty.

- Overall, this project is a simple yet effective solution that saves time, improves accuracy, and supports better academic management in colleges.

Thank You